

Machine Learning Assignment 1

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1 Problem 1. Ridge Regression

Given $\mathbf{x} \in \mathbb{R}^D$, $y \in \mathbb{R}$ and a model $f(\mathbf{x}) : \mathbb{R}^D \rightarrow \mathbb{R}$. We suppose to have a training dataset $\{\Phi, \mathbf{t}\}$ where $\Phi_{N \times D}$ is the design matrix in which each row is a feature vector $\phi(\mathbf{x})$ of a training point $\mathbf{x}$, $\mathbf{t}_{N \times 1}$ is the vector with target values for the training points. N is the number of points in the training dataset, D is the dimensionality of the feature space.

We want to derive the closed form solution for the optimal parameters of a ridge regression model:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}).$$

The optimal parameters give the minimum to the following loss function:

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (f(x_n) - t_n)^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

where $\|\mathbf{w}\|^2 = \mathbf{w}^T \mathbf{w}$.

Solution:

$$\frac{\partial E}{\partial w_j} = \sum_{n=1}^N (f(x_n) - t_n) \frac{\partial f(x_n)}{\partial w_j} = \sum_{n=1}^N \left(\mathbf{w}^T \phi(x_n) - t_n \right) \phi_j(x_n) + \lambda w_j^T$$

In vectorial form it becomes:

$$\frac{dE}{d\mathbf{w}} = \left(\mathbf{w}^T \sum_{n=1}^N \phi(x_n) \phi(x_n)^T - \sum_{n=1}^N t_n \phi(x_n)^T \right) + \lambda \mathbf{w}^T.$$

Introducing

$$\Phi = \begin{pmatrix} \phi_0(x_1) & \dots & \phi_D(x_1) \\ \dots & \dots & \dots \\ \phi_0(x_N) & \dots & \phi_D(x_N) \end{pmatrix} \quad \text{and} \quad \mathcal{T}^T = (t_1, \dots, t_N)^T$$

$$\frac{dE}{d\mathbf{w}} = \mathbf{w}^T \Phi^T \Phi - \mathcal{T}^T \Phi + \lambda \mathbf{w}^T$$

Setting it to zero and transposing:

$$\Phi^T \Phi \mathbf{w} - \Phi^T \mathcal{T} + \lambda \mathbf{w} = 0$$

$$\mathbf{w} \left(\Phi^T \Phi + \lambda \mathbb{1} \right) = \Phi^T \mathcal{T}$$

Finally we get the solution:

$$\mathbf{w}_{\min} = \left(\Phi^T \Phi + \lambda \mathbb{1} \right)^{-1} \Phi^T \mathcal{T}$$

2 Problem 2. Feature engineering

We are given the following set of points in Figure 1.

We want to find new features, such that in the new feature space, classes C1 and C2 are linearly separable. We use the two following features:

$$\phi_1(x, y) = x^2 + y^2 \quad \phi_2(x, y) = \frac{1}{\sqrt{x^2 + y^2}} \cdot \sin \left(\sqrt{x^2 + y^2} \right)$$

where ϕ_2 is the so-called "sombbrero" function. They are shown in Figure 2. In particular, we choose the 'sombbrero' function because of its peak at the center of the (x,y) plane. We thought that this peak can be useful to separate the blue points in Figure 1 that are close to the (0,0) coordinates.

If we use these two features to plot the data points in Figure 1 on the feature space, we obtain the result we were looking for: the two classes become linearly separable, as shown in Figure 3.

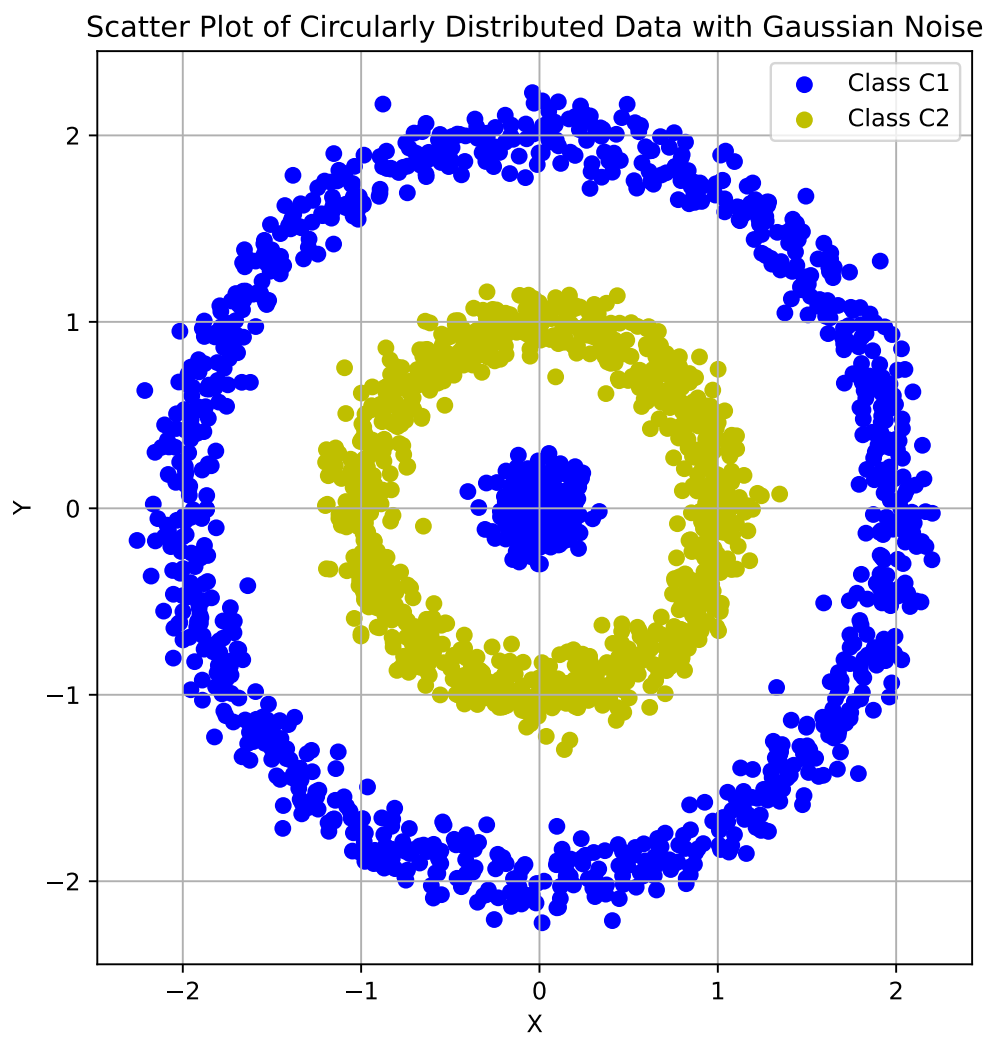


Figure 1

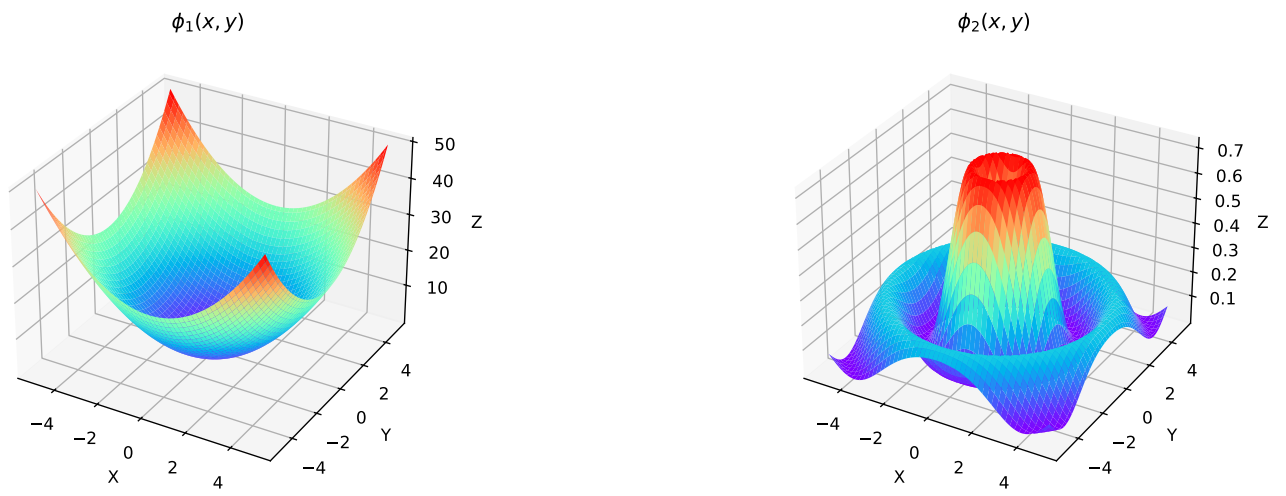


Figure 2

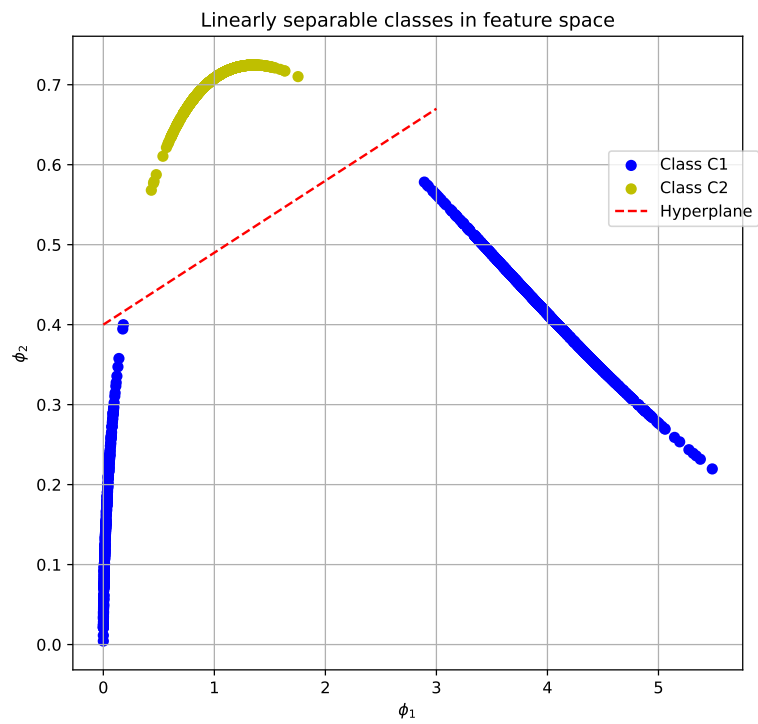


Figure 3

3 Problem 3. Kernel functions

We are given the following function: $f : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$ defined as

$$f(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{x} \mathbf{y}^T \mathbf{y}$$

We want to prove that it is a valid kernel.

Solution:

If for a function $k(\mathbf{x}_n, \mathbf{x}_m)$ the Gram matrix $\mathbf{K} = (k(\mathbf{x}_n, \mathbf{x}_m))_{n,m=1,\dots,D}$ is positive semi-definite for any finite dataset $(\mathbf{x}_n)_{n=1,\dots,D}$, we can write k as a scalar product in some (possibly infinite-dimensional) feature space:

$$k(\mathbf{x}_n, \mathbf{x}_m) = \phi(\mathbf{x}_n)^T \phi(\mathbf{x}_m)$$

for some mapping function ϕ . In this case k is called valid kernel. So, we want to verify that $\mathbf{K}$ is positive semi-definite. First of all, we observe that f is symmetric, i.e. $f(\mathbf{x}, \mathbf{y}) = f(\mathbf{y}, \mathbf{x})$. Therefore, $\mathbf{K}$ is a $D \times D$ symmetric matrix: $\mathbf{K} = (f(\mathbf{x}_n, \mathbf{x}_m))_{n,m=1,\dots,D}$. We now check its positive semi-definiteness, i.e. if $\forall \mathbf{c} \in \mathbb{R}^D \quad \mathbf{c}^T \mathbf{K} \mathbf{c} \geq 0$.

$$\begin{aligned} \sum_{i=1}^D \sum_{j=1}^D c_i K_{ij} c_j &= \sum_{i=1}^D \sum_{j=1}^D c_i (x_i^T x_i x_j^T x_j) c_j = \\ &= \sum_{i=1}^D \sum_{j=1}^D (x_i^T x_i) c_i x_i^T x_j c_j (x_j^T x_j) = \\ &= \sum_{i=1}^D \sum_{j=1}^D \|x_i\|^2 c_i \langle x_i, x_j \rangle \|x_j\|^2 = \\ &= \|\mathbf{x}\|^2 \langle \sum_i c_i x_i, \sum_j c_j x_j \rangle \|\mathbf{x}\|^2 \end{aligned}$$

if we define $y = \sum_i c_i x_i$, then $\langle y, y \rangle \geq 0$. Since also $\|\mathbf{x}\|^2 \geq 0$, we conclude that $\mathbf{K}$ is positive semi-definite and f is a valid kernel.

4 Problem 4. SVM

We are given the training points in Figure 4 and we observe that they are divided into two linearly separable classes. We then construct the weight vector of the maximum

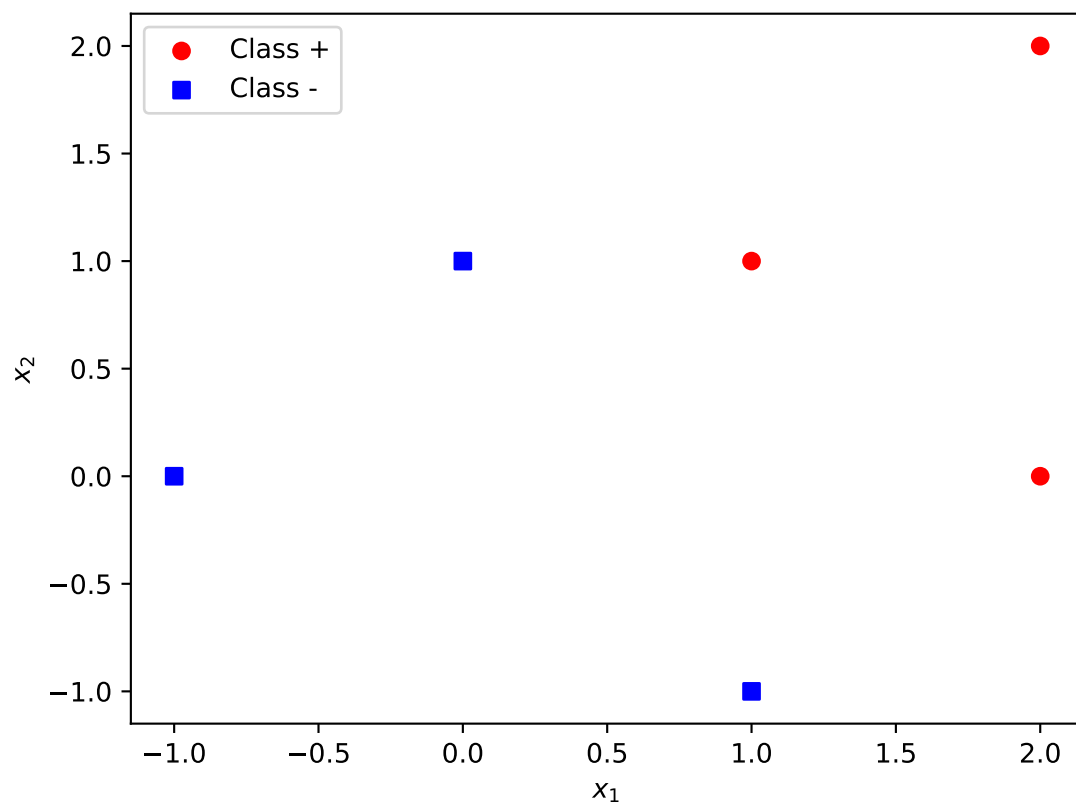


Figure 4

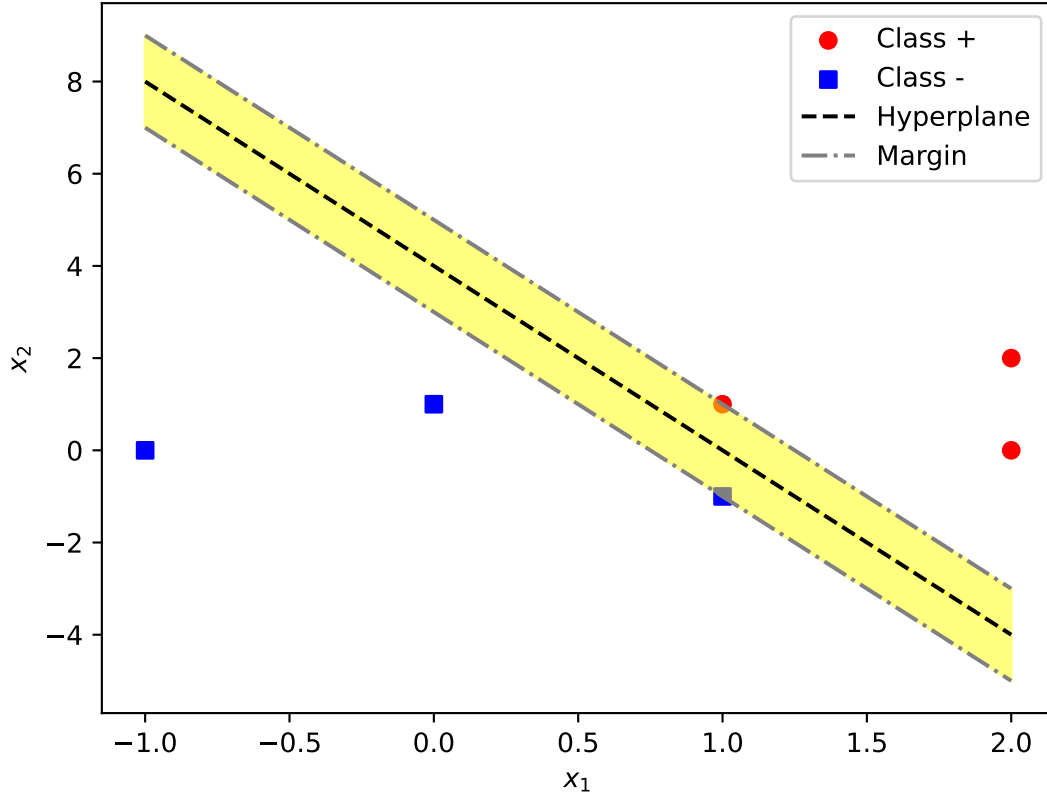


Figure 5

margin hyperplane by inspection as shown in Figure 5. This procedure is implemented in file `SVM.py`. By inspection we found the parameters of the hyperplane:

$$y = mx + q \quad m = -4, \quad q = 4$$

Then we verified that the two support vectors (1,-1) and (1,1) are equidistant from the hyperplane. In particular we computed their distance from the hyperplane applying the following formula:

$$d = \frac{|-mx_1 + x_2 - q|}{\sqrt{m^2 + 1}}$$

Finally, we defined the margin of the linear classifier as twice the distance from the decision boundary to these two nearest point (of either class). In Figure 5 we show these results graphically.

If we remove one of the two support vectors, the size of the optimal margin does not change since the two vectors are equidistant from the classifier. This is not true in general: if the margin is defined just by one support vector (the closest to the hyperplane), when we remove it the size of the margin increases and it is defined as twice the distance from the decision boundary to the second nearest point.